

Single Inheritance Questions

1. Student Management System (Easy)

Parent Class

Student

Attributes:

- student_name
- roll_no

Method:

- display_student()

Child Class

CollegeStudent

Attribute:

- course_name

Method:

- display_course()

Requirement:

Use super().

2. Vehicle Information (Easy)

Parent Class

Vehicle

Attributes:

- brand
- model

Method:

- show_vehicle()

Child Class

Bike

Attribute:

- mileage

Method:

- show_bike()

Requirement:

Do NOT use super().

3. Employee Management (Medium)

Parent Class

Employee

Attributes:

- employee_name
- salary

Method:

- display_employee()

Child Class

Developer

Attribute:

- programming_language

Method:

- display_developer()

Requirement:

Use super().

4. Bank System with Abstraction (Medium)

Abstract Class

Bank

Attributes:

- account_holder
- account_number

Abstract Method:

- interest_rate()

Child Class

SBI

Implement:

- interest_rate()

Requirement:

Use super().

5. Animal Information (Easy)

Parent Class

Animal

Attribute:

- animal_name

Method:

- eat()

Child Class

Dog

Method:

- bark()

Requirement:

No super().

Multilevel Inheritance Questions

6. School Management System (Easy)

Grand Parent

Person

Attribute:

- name

Parent

Student

Attribute:

- roll_no

Child

CollegeStudent

Attribute:

- course_name

Display all details.

Requirement:

Use super().

7. Company Hierarchy (Medium)

Grand Parent

Company

Attribute:

- company_name

Parent

Department

Attribute:

- department_name

Child

Employee

Attribute:

- employee_name

Display all information.

Requirement:

Use super().

8. Banking System (Medium)

Abstract Class

Bank

Attribute:

- bank_name

Abstract Method:

- show_interest()

Parent

SBI

Attribute:

- branch_name

Child

Customer

Attribute:

- customer_name

Display all details.

Requirement:

Use Abstraction + Multilevel Inheritance + super().

9. E-Commerce System (Medium)**Grand Parent**

Product

Attribute:

- product_name

Parent

Electronics

Attribute:

- warranty

Child

Mobile

Attribute:

- price

Display all details.

Requirement:

Without super().

10. Hospital Management System (Hard)**Abstract Class**

Hospital

Attribute:

- hospital_name

Abstract Method:

- show_specialization()

Parent

Doctor

Attribute:

- doctor_name

Child

Cardiologist

Attribute:

- experience

Methods:

- show_all_details()

Requirement:

- Abstraction
 - Multilevel Inheritance
 - super()
 - Method Overriding
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Bonus Interview Question

Create:

Parent Class

Person

Attributes:

- name
- age

Child Class

Teacher

Attributes:

- subject

Requirements

- Use super()
- Add method update_subject()
- Display updated information